JC2001 Introduction to Software Engineering

Practical Session 8 - Exercises

(Model Answers)

Design Patterns

Exercise A: Pattern Description

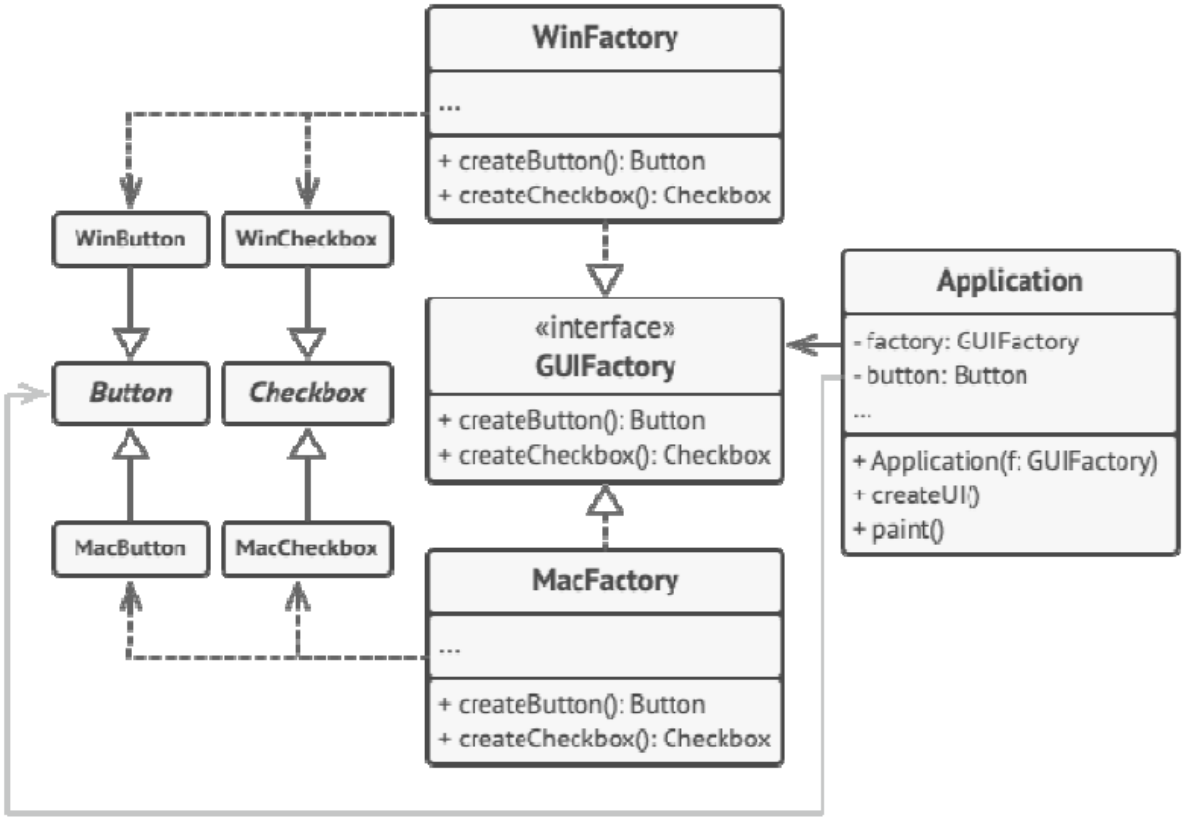
Question 1: What is the main difference between a design pattern and an algorithm?
Answer: Design patterns and algorithms describe typical solutions to known problems. An algorithm is comprised of a clearly defined set of instructions for achieving some goal. In contrast, a design pattern is a high-level description of a solution. It provides a general guideline for how to go about writing and organising a piece of code.

Question 2: Describe the benefits of design patterns in software development.
Answer: Design patterns help promote easier program changes and code reusability. Loosely coupled objects are easier to reuse and change. Design patterns help you write code faster by providing a clearer picture of how you are implementing a design. Design patterns help you analyse the more abstract areas of a program by providing concrete, well-tested solutions.

Exercise B: Creational Patterns

Question 3: Describe the advantages and disadvantages of the Abstract factory pattern.
Answer: Advantages are (a) products you are getting from a factory are compatible with each other, (b) avoid tight coupling between concrete products and client code, (c) you can extract the product code into one place, making the code easier to support and (d) you can introduce new variants of products without breaking existing client code. Disadvantage is the code may become more complicated than it should be, since a lot of new interfaces and classes are introduced along with the pattern.

Question 4: You are a user-interface (UI) engineer in EFG software house. As part of your project, you will be implementing a user friendly graphical user interface. The client specifically asked for a crossplatform GUI, i.e., the GUI must work on Microsoft Windows, Macbook and Linux operating systems. The UI consists of buttons, checkboxes and drop boxes. Identify an appropriate design pattern and draw a class diagram for the cross-platform GUI.
Answer: Abstract factory pattern:


WinFactory
...
+ createButton(): Button
+ createCheckbox(): Checkbox
WinButton
WinCheckbox
«interface»
GUIFactory
+ createButton(): Button
+ createCheckbox(): Checkbox
Application
- factory: GUIFactory
- button: Button
...
+ Application(f: GUIFactory)
+ createUI()
+ paint()
Button
Checkbox
MacButton
MacCheckbox
MacFactory
...
+ createButton(): Button
+ createCheckbox(): Checkbox